2.7.1 Extension to Multiple Inputs

- A gate can be extended to have multiple inputs if the binary operation it represents is commutative and associative.
- The AND and OR operations, defined in Boolean algebra, possess these two properties.

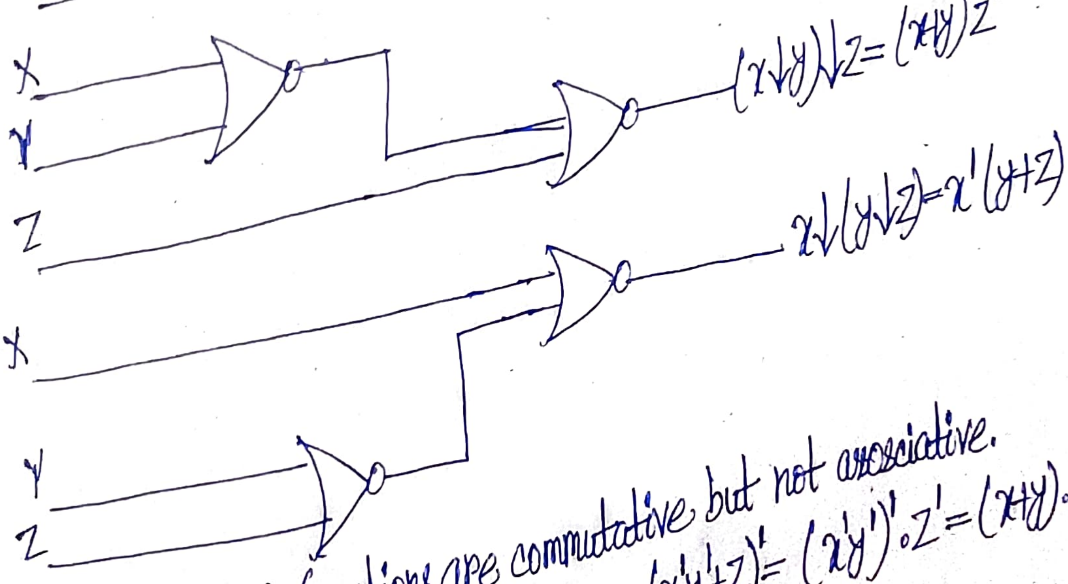
For the OR function we have:

$$\boxed{x+y=y+x} \quad \text{commutative}$$

and $(x+y)+z = x+(y+z) = x+y+z$ associative

which indicates that the gate inputs can be interchanged and that the OR function can be extended to three or more variables.

- Figure 2.6 Demonstrating the nonassociativity of the NOR operator; $(x \downarrow y) \downarrow z \neq x \downarrow (y \downarrow z)$



- NAND and NOR functions are commutative but not associative.

$(x \downarrow y) \downarrow z = (x+y)' \downarrow z = (x'y') \downarrow z = (x'y'+z)' = (x'y')' \cdot z' = (x+y) \cdot z'$
$= xz' + yz'$

$x \downarrow (y \downarrow z) = x \downarrow (y+z)' = x \downarrow (y'z') = (x+y'z')' = x' \cdot (y'z')' = x'(y+z)$

To overcome this difficulty, we define the multiple NOR (or NAND) gate as a complemented OR (or AND) gate. Thus, by definition, we have:

$$\boxed{\begin{array}{l} x \downarrow y \downarrow z = (x+y+z)' \\ x \uparrow y \uparrow z = (xyz)' \end{array}}$$